National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Electronic Devices & Circuits	Course Code:	EE1004
	Program:	Electrical Engineering	Semester:	Fall2021
	Duration:	180 Minutes	Total Marks:	100
	Paper Date:	07-01-2022	Weight	50%
	Section:	ALL	Page(s):	16
	Exam Type:	Final		

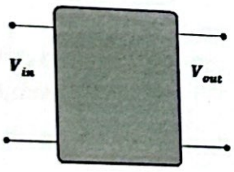
Student : Name:_______________ Roll No._______________ Section:________

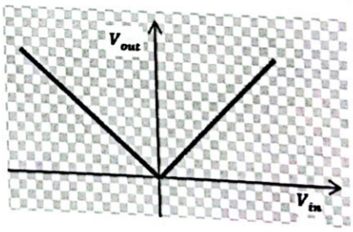
Instruction/Notes: Write your roll number on each page.
Carefully read the paper and attempt each question in the space provided only.
Show all calculations, direct answers are not acceptable. Your answers should be approximated up to two decimal places.
State your valid assumptions clearly if you have to take any.

	Q1 (CLO#1)	Q2 (CLO#2)	Q 3 (CLO#3)	Q4 (CLO#3)	Q5 (CLO#4)	Total
Total marks	20	15	25	15	25	100
Marks obtained						
CLO score						

Question: 1 (CLO # 1) **20 Marks**

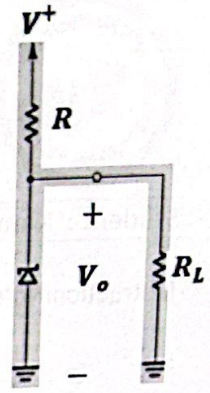
a) **Draw the circuit inside the black box which has the transfer characteristics as given below.**





b) The 5.1-V Zener diode in the circuit is specified to have $V_Z = 5.1\ V$ at $I_Z = 4\ mA$, $r_Z = 25\ \Omega$, $R = 2K\Omega$ and $I_{ZK} = 0.3\ mA$. The supply voltage V^+ is nominally 15 V but can vary by $\pm 1\ V$.

i. Find V_o with no load and with V^+ at its nominal value



ii. Find the line regulation.

iii. Find the load regulation if $R_L = 3\ k\Omega$.

iv. What is the minimum value of R_L for which the diode still operates in the breakdown region?

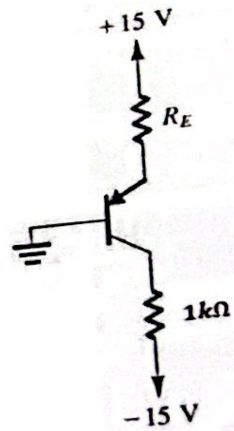
v. Keeping $R_L = 2\ k\Omega$, find the maximum value of R so that diode stays in the breakdown region?

15 Marks

Question: 2 (CLO # 2)

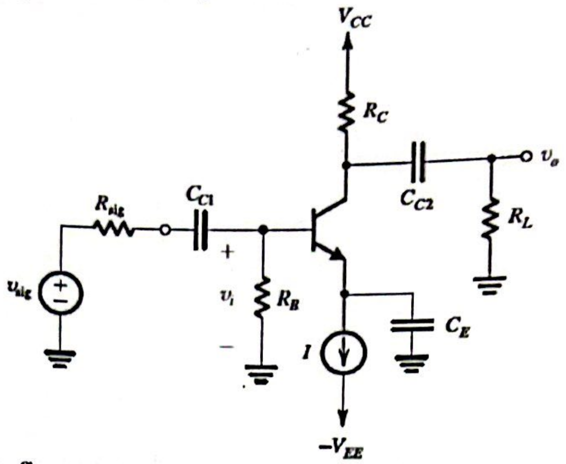
For the following circuit having a BJT with = 100 , select a value of R_E such that

a) BJT saturates with a forced β of 10.
b) BJT is at the edge of saturation.



Question: 3 (CLO # 3) **25 Marks**

Consider the following circuit with $V_{CC} = V_{EE} = 10V, I = 1mA, R_B = 100\ k\Omega,\ R_C = 8\ k\Omega,\ \beta = 100,\ R_{sig} = 4\ k\Omega$ and $R_L = 4\ k\Omega$.



a) Identify the configuration of the amplifier. Find all dc currents and voltages.

b) **Draw small signal equivalent circuit (include R_{sig} and R_L in your model) .**

c) Write the mathematical expressions and compute the value of input resistance, output resistance and the overall voltage gain G_V.

d) **The designer of the amplifier decides to lower the bias current to half of its original value but maintain the overall voltage gain G_V to its original value. In order to do that, designer changes the value of R_C. For the new design, determine the value of R_C and input resistance R_{in}.**

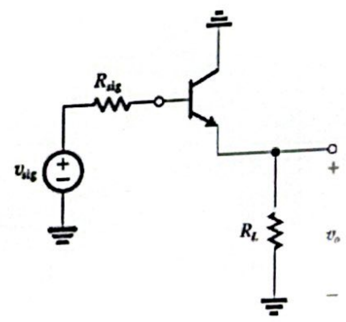
c) **What is the effect of bias current on the input resistance of the amplifier. Explain your results from the calculations of above parts.**

15 Marks

Question: 4 (CLO # 3)

An emitter follower is required to deliver a 0.5V peak sinusoid at the output to a $2K\Omega$ load.

i. **Draw and label small signal equivalent circuit using T-model (include R_{sig} and R_L) .**



ii. If the peak amplitude of v_{be} is to be limited to $5mV$, what is the value of I_E at which the BJT can be biased?

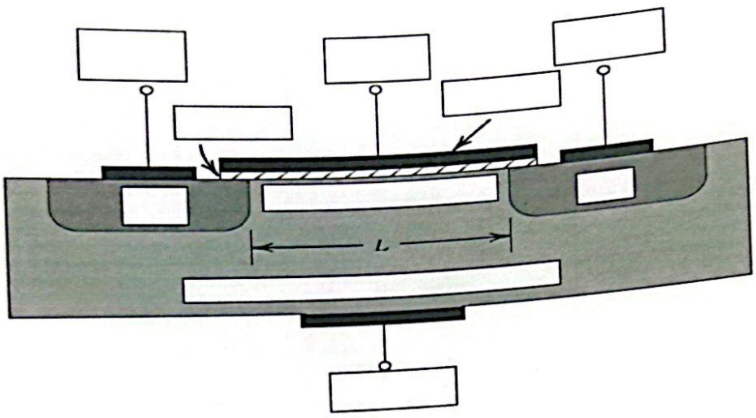
iii. If the resistance of the signal source is $200K\Omega$, what value of G_v is obtained? Also determine the required amplitude of v_{sig}?

iv. Keeping the bias current constant and using the peak value of v_{sig} obtained in part iii, Find the output voltage if $R_L = 1k\Omega$

v. Is there any effect on the voltage gain if we change the load resistance? Explain.

Question: 5 (CLO # 4) **25 Marks**

a) **The physical structure of an N-channel MOSFET is shown below. Label the empty blocks.**

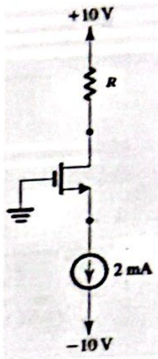


b) **Draw and label $i_D - v_{DS}$ characteristics of an *N*-channel enhancement MOSFET.**
(Show at least 4 curves for 4 different values of V_{GS})
Identify the operating regions of NMOS on the characteristic curve.

c) An NMOS transistor having $V_t = 1V$ is operated in the triode region with small V_{DS}. With $V_{GS} = 1.5V$, it is found to have a resistance r_{DS} of $1k\Omega$.
What value of V_{GS} is required to obtain $r_{DS} = 200\Omega$?
Find the corresponding resistance values obtained with a device having twice the value of Width of channel 'W'?

d) Design the circuit shown in figure, transistor is characterized by

$$k_n' \frac{W}{L} = 1mA/V^2 \text{ and } V_t = 1V, V_D = 2V.$$



e) In the above circuit replace the current source with a resistor and redraw the circuit. Select a value of resistor to yield a current as close to current source as possible.